

University of Florence
Department of Information Engineering (DINFO)

HYDRAS Project Weekly Report

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Velocity sweep 0.1–5 m/s: single vs. double sensor ring

Extending the adaptive-velocity PPO study to the full range

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1 Context and protocol

The PPO policy for source seeking uses an *adaptive-velocity chain*: at each step the agent picks one of $K=5$ discrete velocities in $(0, v_{\max}]$, so the action space is $\text{Discrete}(8 \times 5) = 40$ (identical for every $v_{\max}$, so the weights transfer). This report extends the velocity study to the **whole range** $v_{\max} \in \{0.1, 0.4, 0.7, 1, 1.2, 1.5, 1.7, 1.9, 2, 3, 4, 5\}$ m/s and, for each velocity, compares the **single** sensor ring against the **double** ring (a second ring at 50 m in addition to the local one at 20 m). The special interest is the low-velocity region 0.1–1 m/s, never characterised before.

Models. All single-ring models share the same protocol (fine-tuning, constant $lr=10^{-5}$, $2 \cdot 10^6$ timesteps, oversampling $\alpha=0.8$ of the hard scenarios, success radius 50 m):

- the integer chain $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$ (each stage resumes the previous);
- the intermediate stages 1.2/1.5/1.7/1.9, fine-tuned from v_1 ;
- the low velocities 0.7/0.4/0.1, a **descending** chain $v_1 \rightarrow 0.7 \rightarrow 0.4 \rightarrow 0.1$.

Each **double-ring** model is obtained by **warm-start** from its single-ring counterpart at the same $v_{\max}$: the input layer is widened from 116 to 196 dimensions with the ring-2 columns **initialised to zero**, so at start-up the network is identical to the single ring and fine-tuning decides whether and how to use the second ring. As a training-free baseline, the FCM (Adam) gradient climber is evaluated at $lr = 10 \cdot v_{\max}$ (sensor 50 m). The warm-start is illustrated below (example at $v_{\max}=1$; identical at every level, from the respective single-ring model).

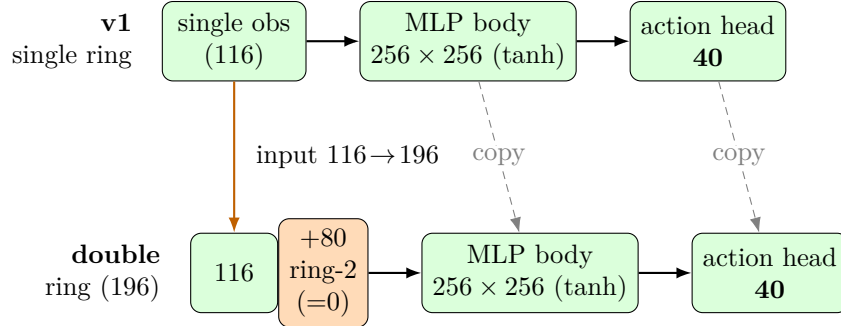


Figure 1: Warm-start of the double ring (example at $v_{\max}=1$: $v_1 \rightarrow$ double ring; it is analogous at every level, from the respective single-ring model). **Body** and **action head** are **copied** unchanged (green); only the **input layer** is widened from 116 to 196 (orange), adding 80 columns for ring-2 with **weights initialized to zero**. At start-up the network is identical to v_1 and ignores ring-2; fine-tuning decides whether and how to use it.

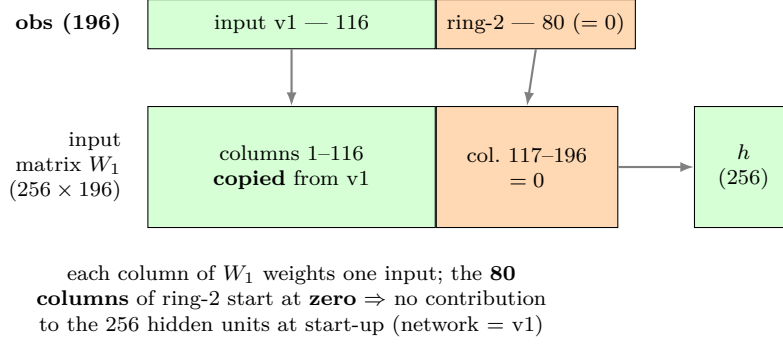


Figure 2: Focus on widening the input layer ($116 \rightarrow 196$). The first-layer matrix gains 80 columns (one per ring-2 sensor), **initialized to zero**: at start-up ring-2 does not affect the hidden units and the network behaves exactly like v1.

Evaluation. Every model is evaluated on the same held-out set: 26 test sources (SRC107–132) $\times$ 4 wind scenarios (V0–V3) $\times$ 3 time chunks (Q1/4, Q1/2, Q3/4) $\times$ 5 episodes = **1560 episodes/model** (FCM: 624), deterministic policy.

2 FCM results

The FCM (Adam) gradient climber is the training-free baseline ($lr = 10 \cdot v_{\max}$, sensor 50 m). Table 1 reports its success rate over the whole velocity range, broken down by wind scenario (V0–V3) and time chunk (Q1/4, Q1/2, Q3/4). The picture is flat and low: the global SR never exceeds $\sim 63\%$, and it collapses in the **adverse-current** winds (V1, V2: 13–42%) and the **later chunks** (Q1/2 as low as 10%), where the plume is advected away from the source and the local gradient no longer points home.

$v_{\max}$	SR%	V0	V1	V2	V3	Q1/4	Q1/2	Q3/4
0.1	29.3	22.4	12.8	19.2	62.8	50.5	9.6	27.9
0.4	48.1	60.9	26.9	34.6	69.9	72.1	23.1	49.0
0.7	54.0	75.0	28.2	35.3	77.6	75.5	32.2	54.3
1.0	54.6	73.7	29.5	35.9	79.5	77.9	30.3	55.8
1.2	56.4	76.3	29.5	37.2	82.7	79.8	34.6	54.8
1.5	57.1	77.6	31.4	38.5	80.8	81.7	30.8	58.7
1.7	57.2	79.5	31.4	34.0	84.0	82.2	37.5	51.9
1.9	58.0	75.0	33.3	37.2	86.5	83.7	37.0	53.4
2.0	58.2	80.1	32.1	34.6	85.9	81.7	38.0	54.8
3.0	60.7	84.6	30.8	37.2	90.4	86.1	39.4	56.7
4.0	63.5	84.6	35.3	41.7	92.3	90.9	44.2	55.3
5.0	62.2	84.6	34.6	37.2	92.3	89.4	41.8	55.3

Table 1: FCM (Adam, $lr = 10 \cdot v_{\max}$, sensor 50 m): global SR and SR per wind scenario (V0–V3) and time chunk (Q1/4, Q1/2, Q3/4), held-out (624 episodes/model). All values are success rate (%). FCM saturates around 63% and stays weak in the advected-plume cells (V1, V2, Q1/2).

3 PPO results: single vs. double ring

$v_{\max}$	SR single	SR double	Δ SR	step single	step double	Δ step
0.1	59.0%	49.0%	−10.0	478.9	423.2	−11.6%
0.4	91.7%	88.5%	−3.2	183.4	187.7	+2.3%
0.7	96.0%	93.9%	−2.1	136.6	131.3	−3.9%
1.0	96.8%	95.8%	−1.0	114.6	97.8	−14.7%
1.2	98.4%	98.7%	+0.3	95.5	76.5	−19.9%
1.5	98.5%	97.3%	−1.2	86.3	64.7	−25.0%
1.7	99.9%	99.6%	−0.3	68.7	60.7	−11.6%
1.9	99.0%	99.9%	+0.9	57.8	47.7	−17.5%
2.0	99.6%	99.7%	+0.1	71.9	49.8	−30.7%
3.0	99.9%	100.0%	+0.1	46.8	31.3	−33.1%
4.0	99.9%	99.9%	0.0	39.7	25.3	−36.3%
5.0	99.9%	100.0%	+0.1	27.4	20.9	−23.7%

Table 2: Single vs. double sensor ring at each $v_{\max}$ (held-out, 1560 episodes/model). “step” = average steps to success (1 step = 10 s). Δ SR = double − single (percentage points); Δ step = relative step change (green = double faster / better, red = double worse).

Two regimes emerge, split around $v_{\max} \approx 1\text{--}1.2$ m/s:

- **Low velocity (0.1–1 m/s): the double ring *costs* SR.** The penalty is largest at the very bottom (−10 points at 0.1, −3.2 at 0.4) and shrinks as velocity grows (−2.1, −1.0). With small steps the agent stays immersed in the field longer, and in the adverse scenarios the far (50 m) signal can point *along the current* rather than toward the source: on distant spawns the agent trusts the far reading, takes the wrong direction and times out. It is the same mechanism already identified at $v_{\max}=1$, amplified as $v_{\max}$ decreases.
- **From $v_{\max} \geq 1.2$: same reliability, and consistently faster.** SR is equal or higher for the double ring (Δ SR between −1.2 and +0.9, i.e. noise near saturation), while the steps to success drop by 12–37%. The value of the “far view” scales with the step length: at higher $v_{\max}$ the strides are large, a badly aimed stride costs an overshoot, and the second ring lets the agent aim the long strides better.

The crossover is clear: **the second ring is a net advantage only from the intermediate velocities up**, where it buys speed at no reliability cost; below ~ 1 m/s it is a liability. Both metrics are plotted, together with FCM, in Section 4.

3.1 Low-velocity detail: success rate per scenario

At the bottom of the range the outcome is dominated by the wind/chunk scenario. Table 3 breaks down the SR of the low-velocity models by wind (V0–V3) and time chunk (Q1/4, Q1/2, Q3/4). Two things stand out: (i) the failures concentrate in the

adverse-current scenarios (V1, V2) and the **later chunks** (Q1/2, Q3/4), where the plume is strongly advected — at 0.1 m/s the single ring already falls to $\sim 40\%$ there; (ii) the **double ring’s cost is largest exactly in those cells** (V1/V2 drop to 29–31% at 0.1 m/s), confirming that the far signal misleads the agent precisely when the current dominates and the small steps leave no room to recover.

model	SR%	V0	V1	V2	V3	Q1/4	Q1/2	Q3/4
single 0.1	59.0	75.4	39.7	41.8	79.2	79.0	53.3	44.8
single 0.4	91.7	98.2	93.3	77.2	98.2	100.0	86.5	88.7
single 0.7	96.0	98.7	100.0	86.2	99.2	100.0	97.1	91.0
double 0.1	49.0	52.6	30.5	29.2	83.6	71.2	38.7	37.1
double 0.4	88.5	96.2	94.6	68.7	94.6	100.0	83.5	82.1
double 0.7	93.9	94.1	100.0	81.5	100.0	100.0	92.1	89.6

Table 3: Low-velocity models: global SR and SR per wind scenario (V0–V3) and time chunk (Q1/4, Q1/2, Q3/4), held-out (1560 episodes/model). All values are success rate (%). The failures — and the double-ring penalty — concentrate in V1/V2 and Q1/2, Q3/4.

4 FCM vs. PPO

Figures 3 and 4 put the two methods side by side across the velocity range, on the two metrics — global success rate and mean steps to success — Table 4 closes with the head-to-head SR gap.

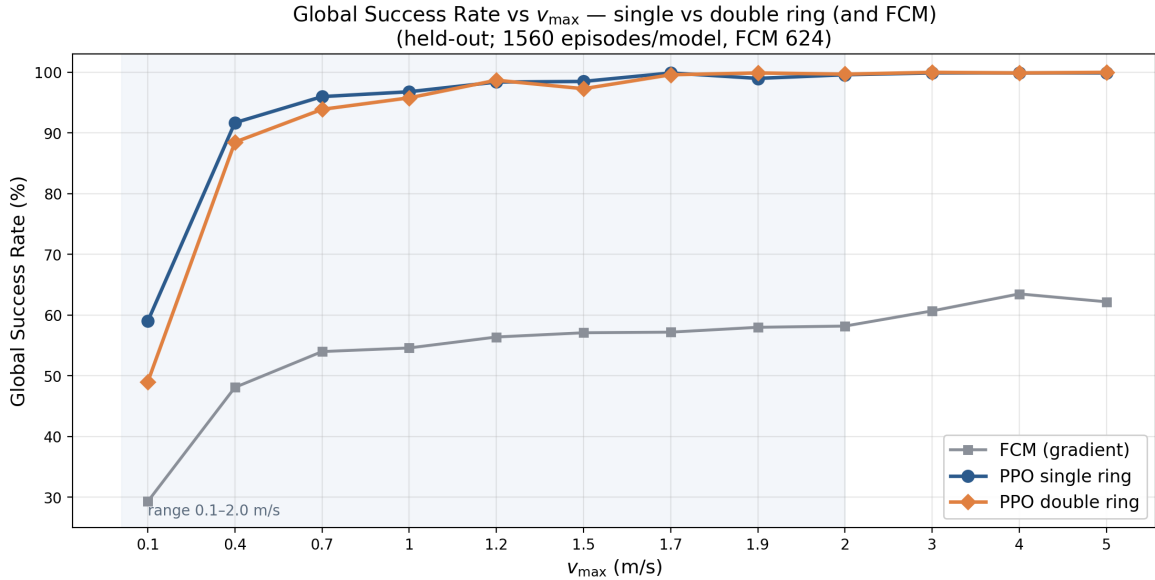


Figure 3: Global success rate vs. $v_{\max}$: PPO single ring, PPO double ring and the FCM baseline. The two PPO curves overlap from ~ 1.2 m/s and saturate near 100%; below 1 m/s the single ring is above the double one. FCM stays far lower at every velocity.

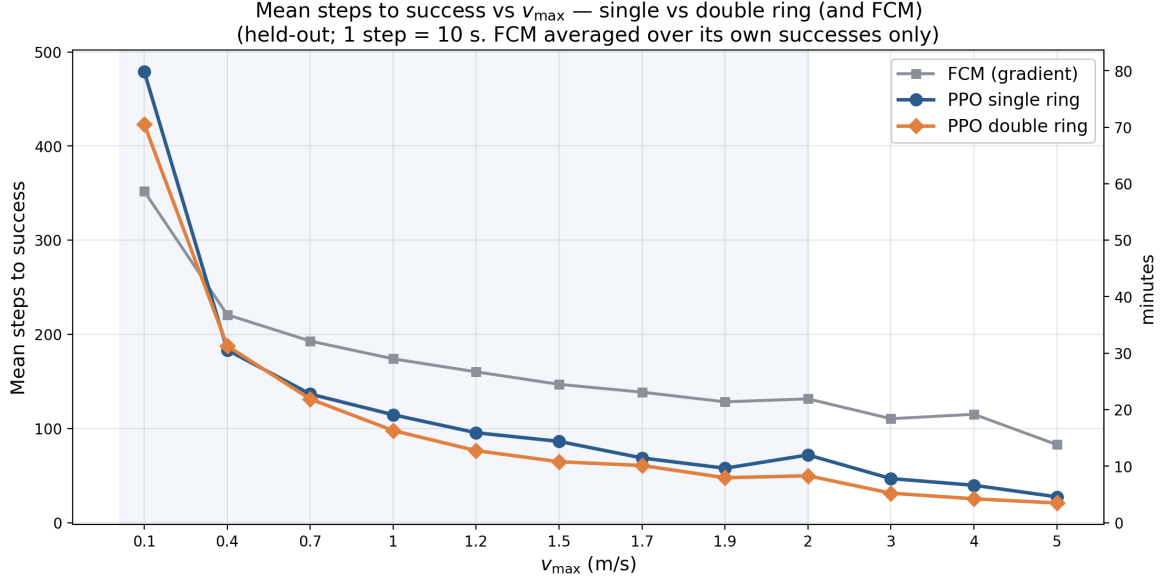


Figure 4: Mean steps to success vs. $v_{\max}$ (right axis in minutes). The double ring is consistently faster than the single from $v_{\max} \geq 1.2$. FCM is shown for reference: its steps are averaged over its *own* successful episodes (the easy ones), so the comparison with PPO is indicative, not at equal episodes. The small bump at $v_{\max}=2$ (single) reflects the mild non-monotonicity between the integer chain and the intermediate stages, trained as independent runs.

The SR gap between the learned policy and the gradient climber is wide and systematic at *every* velocity, and it is largest exactly where the pollutant plume is hardest to follow.

$v_{\max}$ (m/s)	FCM SR	PPO SR	PPO – FCM
0.1	29.3%	59.0%	+29.7
0.4	48.1%	91.7%	+43.6
0.7	54.0%	96.0%	+42.0
1.0	54.6%	96.8%	+42.2
1.2	56.4%	98.4%	+42.0
1.5	57.1%	98.5%	+41.4
1.7	57.2%	99.9%	+42.7
1.9	58.0%	99.0%	+41.0
2.0	58.2%	99.6%	+41.4
3.0	60.7%	99.9%	+39.2
4.0	63.5%	99.9%	+36.4
5.0	62.2%	99.9%	+37.7

Table 4: FCM (Adam, $lr = 10 \cdot v_{\max}$) vs. PPO single ring, global SR at equal maximum velocity (held-out; FCM 624 episodes, PPO 1560). PPO leads by +36–44 points throughout; the FCM saturates around 63% because local gradient ascent, without episode memory or hydrodynamic context, is misled by the advected plume.

5 Conclusions

Extending the velocity study to the full range 0.1–5 m/s, evaluated held-out on 1560 episodes per model, gives a coherent picture on three axes.

1. **Velocity.** Success rate rises steeply with $v_{\max}$ and saturates ($\geq 99\%$) from ~ 1.7 m/s; the steps to success keep dropping up to 5 m/s. The weakness of the low-velocity regime (SR 59% at 0.1, 92% at 0.4) is intrinsic: with tiny steps the agent cannot beat the current in the adverse scenarios.
2. **Single vs. double ring.** The second ring is a **net advantage** from $v_{\max} \geq 1.2$ — same reliability and 12–37% fewer steps — but a **liability below ~ 1 m/s**, where the far signal misleads the agent on distant spawns (up to -10 points of SR at 0.1). The crossover sits around 1–1.2 m/s.
3. **Learned vs. gradient.** PPO dominates the FCM baseline by +36–44 points of SR at every velocity, confirming that a learned policy with memory and context far outperforms pure local gradient ascent.